

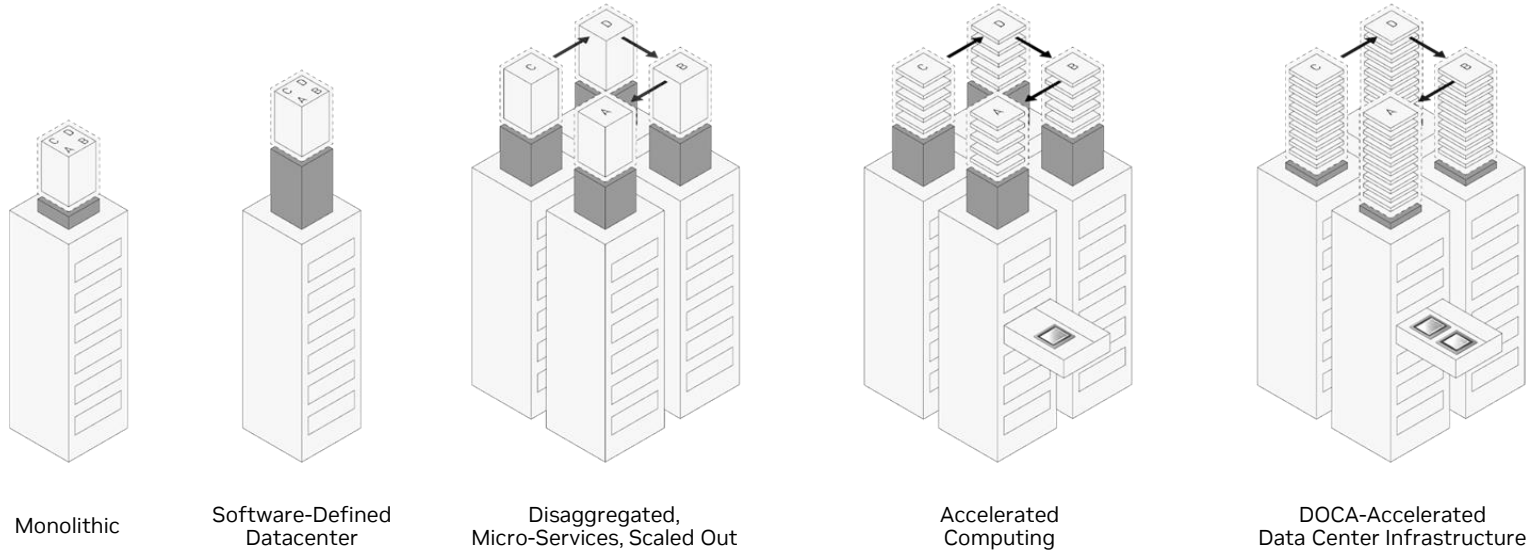


# **Build Next-Generation AI Factories with DOCA-Accelerated Networking**

NVIDIA & Google Cloud | NVIDIA GTC 2025

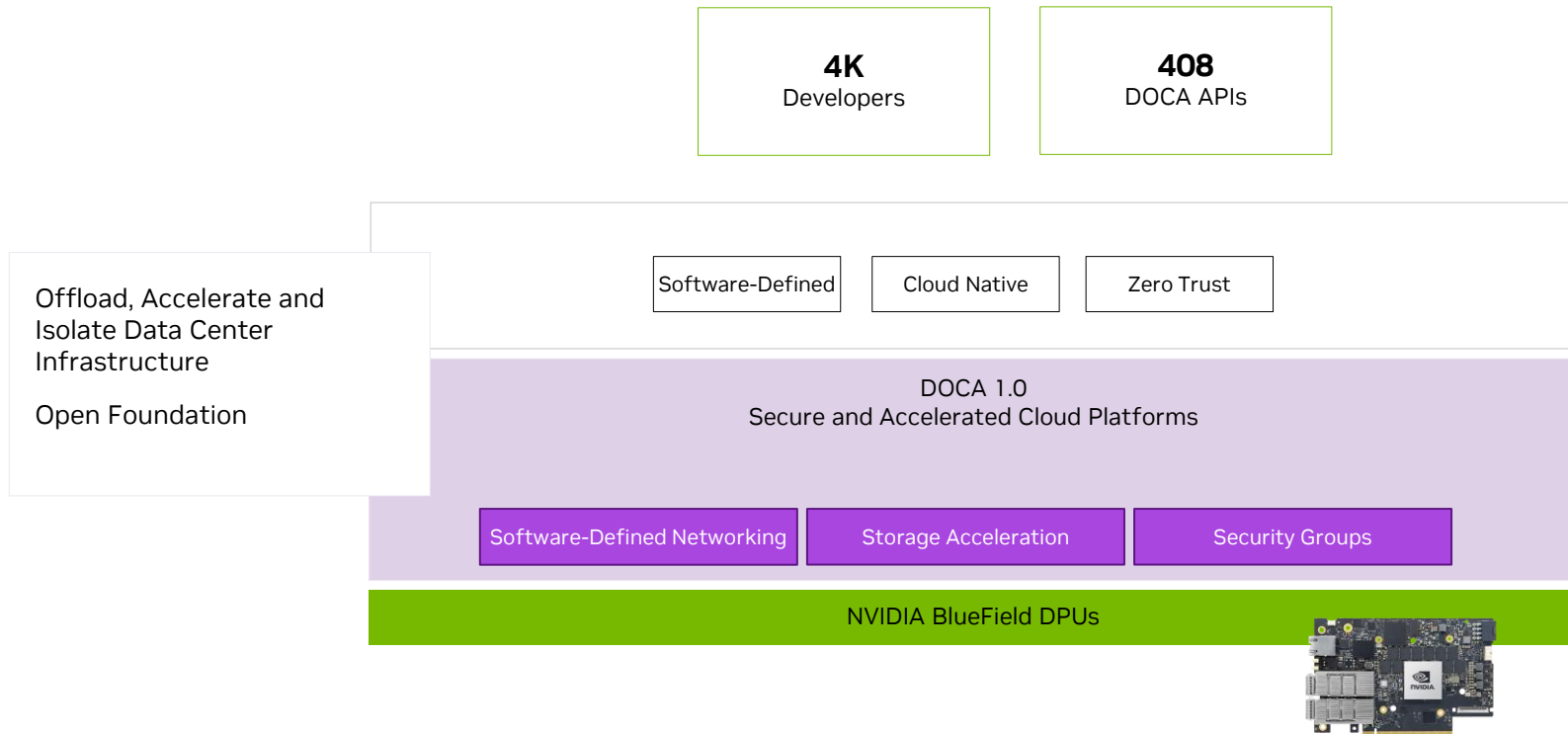
# Transforming Data Centers for the Era of AI

NVIDIA launches DOCA to accelerate data center-scale computing



# NVIDIA DOCA 1.0

Empowering infrastructure developers with a comprehensive software acceleration framework



# NVIDIA DOCA 2.0

Accelerated cloud networking with enhanced programmable I/O

**17K**  
Developers

**1781**  
DOCA APIs

Unified Accelerated  
Framework for Data Center  
Infrastructure

Developers' One Stop Shop

Accelerate the AI Fabric

Software Defined

Cloud Native

Zero Trust

VPC Networking

AI Compute

DOCA 2.0  
Cloud Compute and AI Factory

Programmable Congestion  
Control

Crypto

Telemetry

Software-Defined Networking

Storage Acceleration

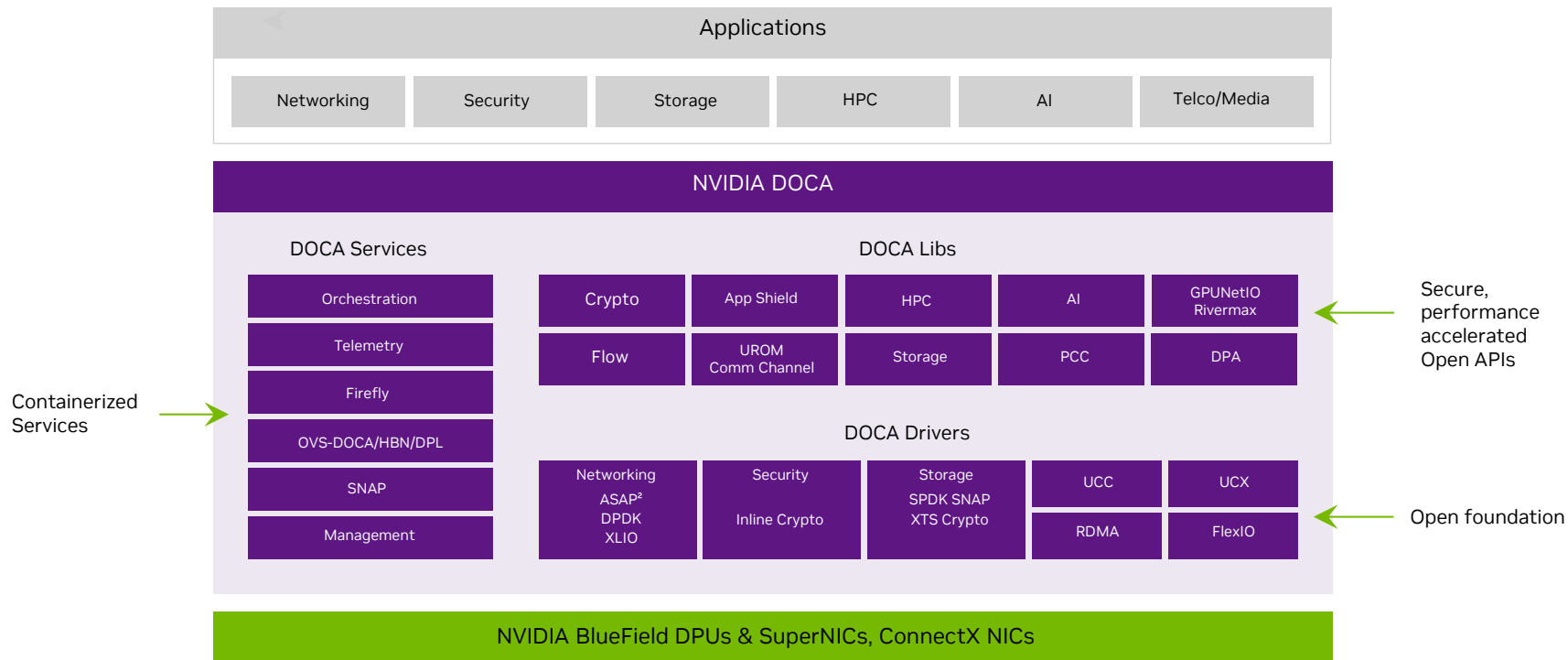
Security Groups

NVIDIA BlueField DPUs & SuperNICs, ConnectX NICs



# NVIDIA DOCA 2.0

Comprehensive acceleration framework, SDK, compilers, services, and tools

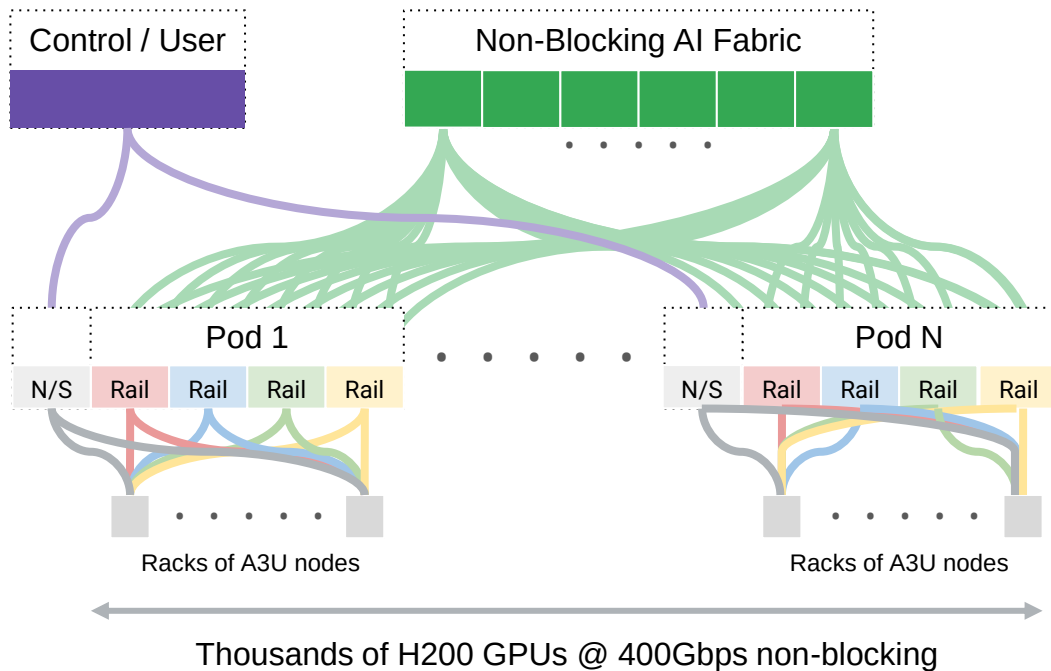


# **DOCA Use Case: AI Hypercomputer**

# A3 Ultra cluster scale-out networking

AI Hypercomputer is a Cloud  
supercomputer architecture.

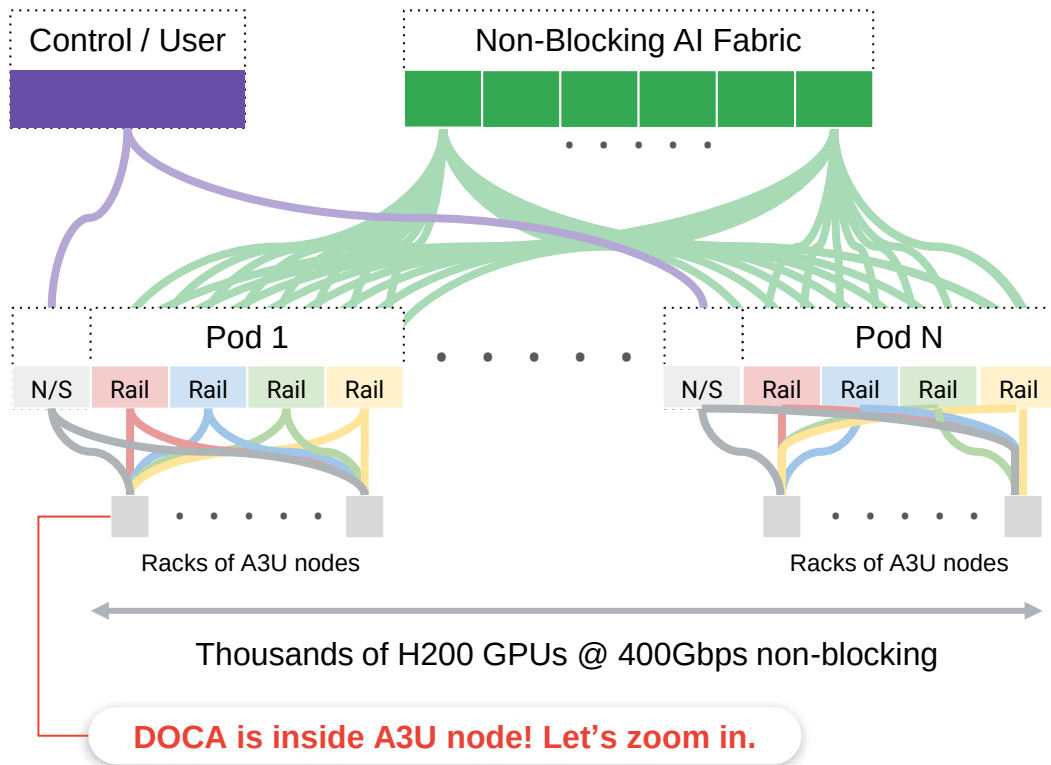
It integrates software, hardware,  
and consumption models to use  
thousands of networked GPUs as  
a single, powerful computer.



# A3 Ultra cluster scale-out networking

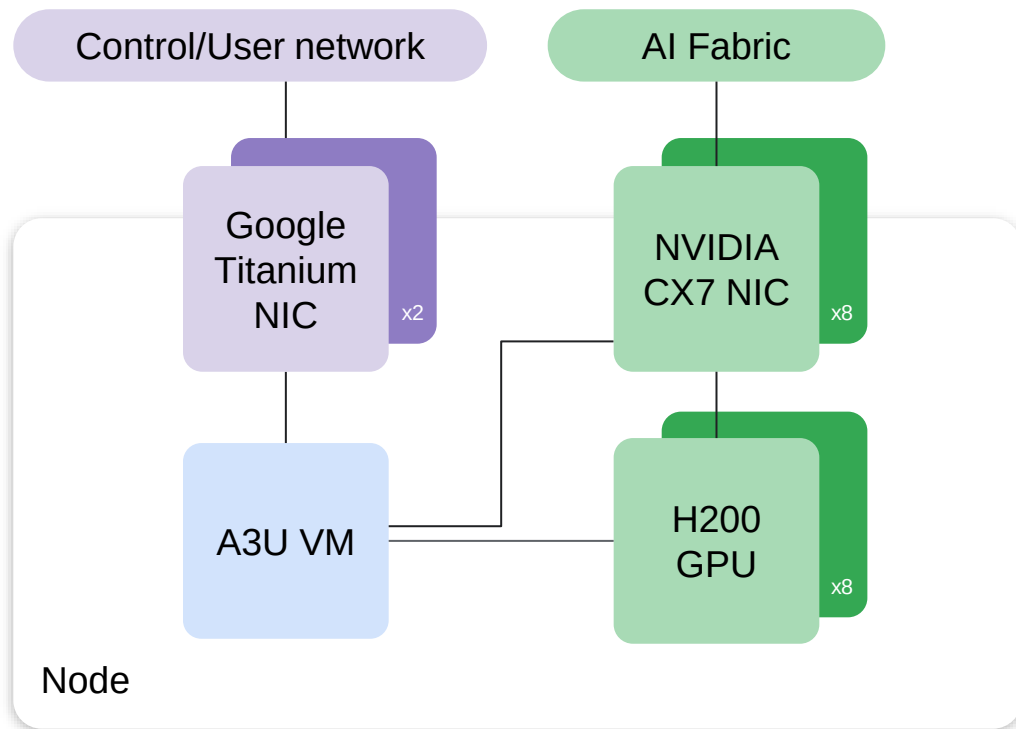
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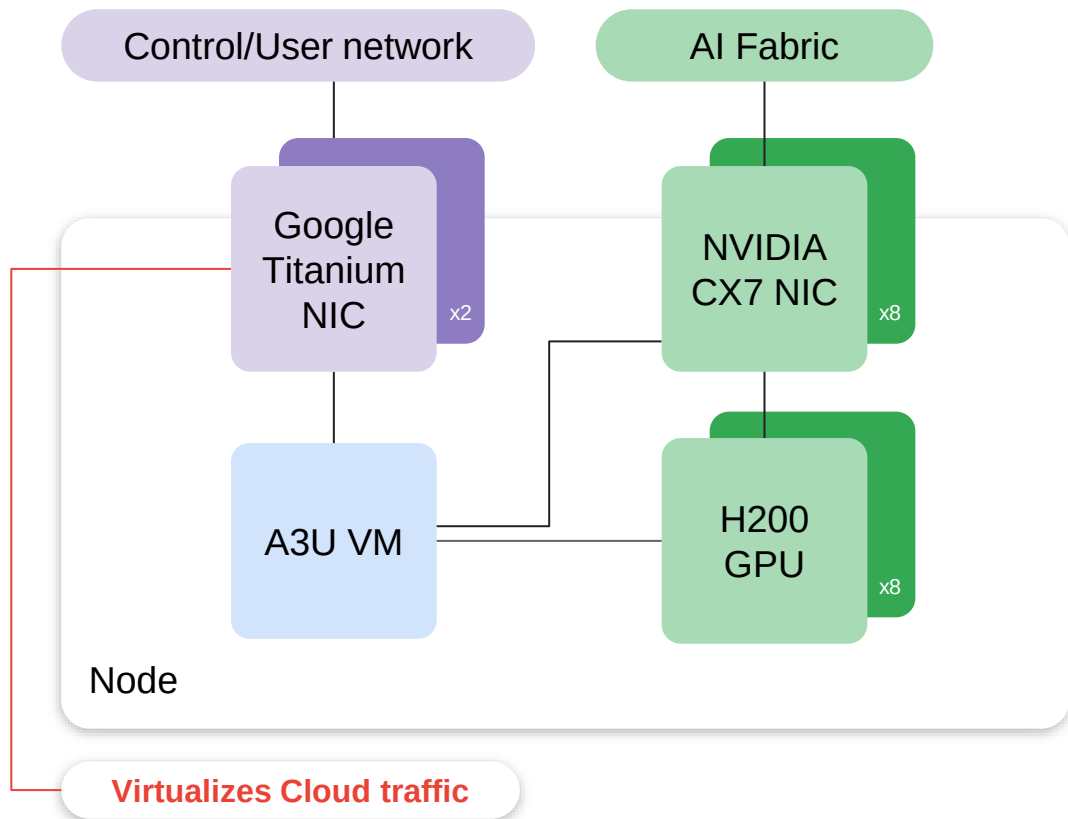




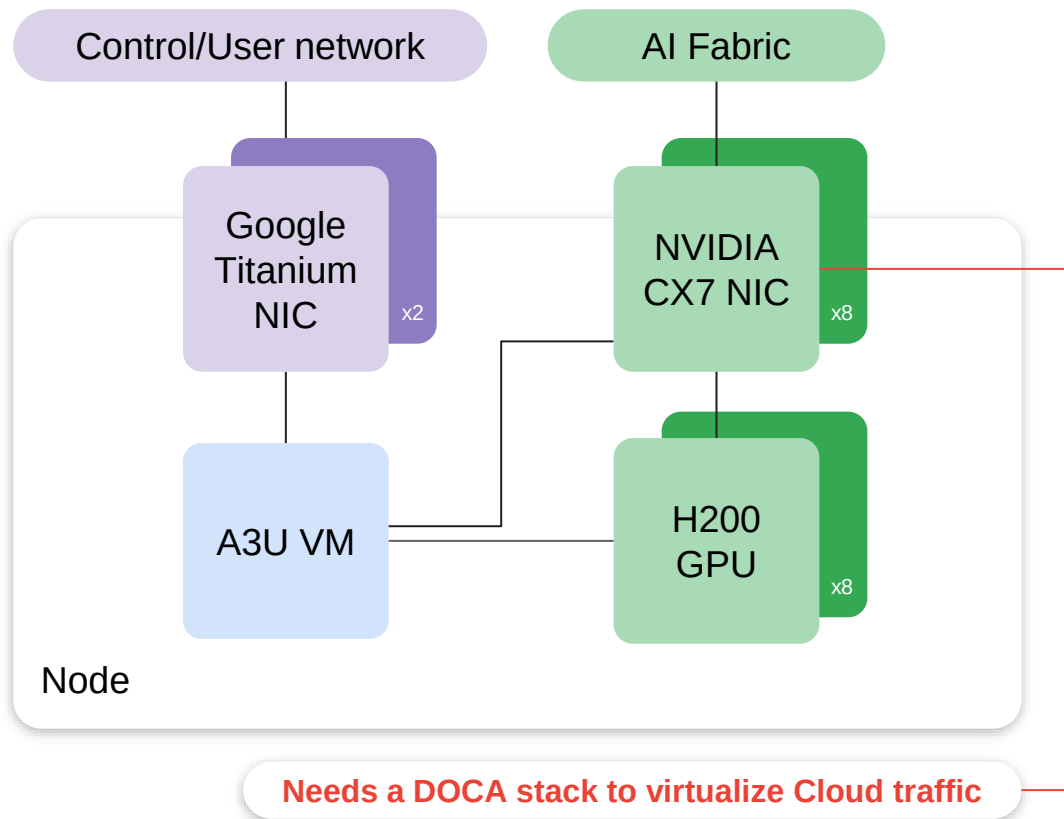
# A3 Ultra VM



# A3 Ultra VM



# A3 Ultra VM



# Google virtualization needs for CX7

## Subset of Titanium virtualization

Parts suitable for GPU-to-GPU traffic

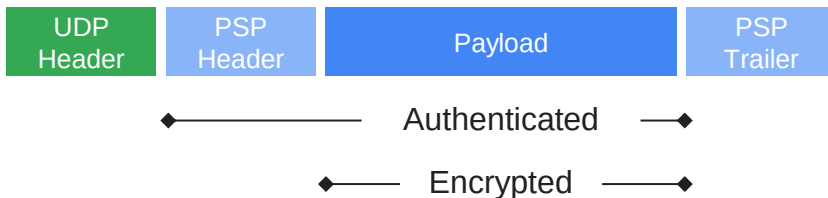
## Multi-tenant isolation is the primary need

Let multiple customers safely use the shared network

## Encapsulation is the primary mechanism

Used by Cloud control plane to restrict reachability

PSP end-to-end encryption assures confidentiality and integrity



# Google virtualization needs (cont.)

## Telemetry & troubleshooting

- Accounting
- Monitoring
- Performance tuning
- Tracing

## In-service software updates (or hitless updates)

Upgrade without disrupting customer

## Cloud networking features

Example: VPC firewalls

Virtualization needs to be implemented on the trusted Node, not in the untrusted VM.

# Using DOCA at Google



## DOCA Flow SDK & CX7 NIC

Basics of DOCA Flow SDK

PSP encryption & VPC overlay  
encapsulation @ Google



## DOCA use cases

Google VPC Traceroute

Hitless Updates

# DOCA Flow SDK & CX7 NIC

## CX7 device

PF and VF device functions.



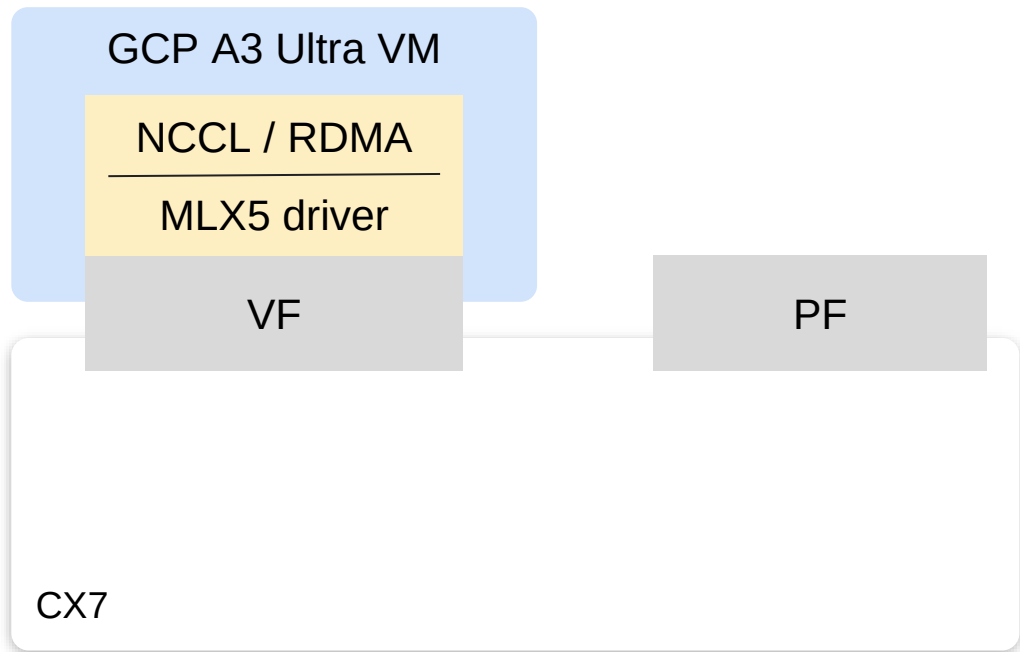
# DOCA Flow SDK & CX7 NIC

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## VMs use CX7 VF function

Customer experience is similar to on-prem deployments.





# DOCA Flow SDK & CX7 NIC

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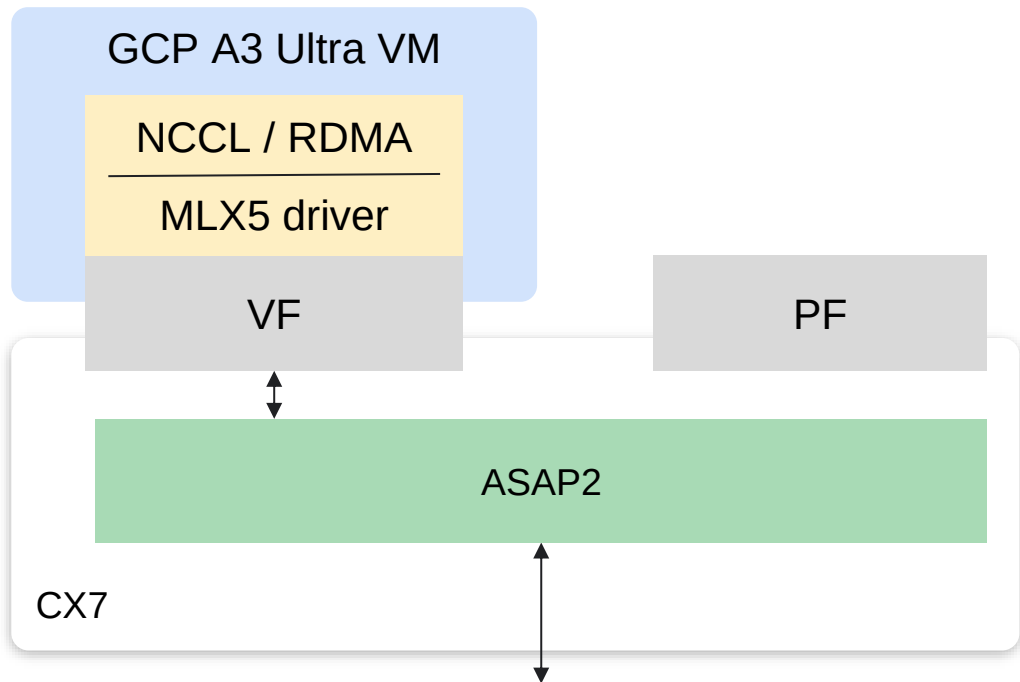
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## CX7 ASAP2 hardware

Processes all the VM packets.



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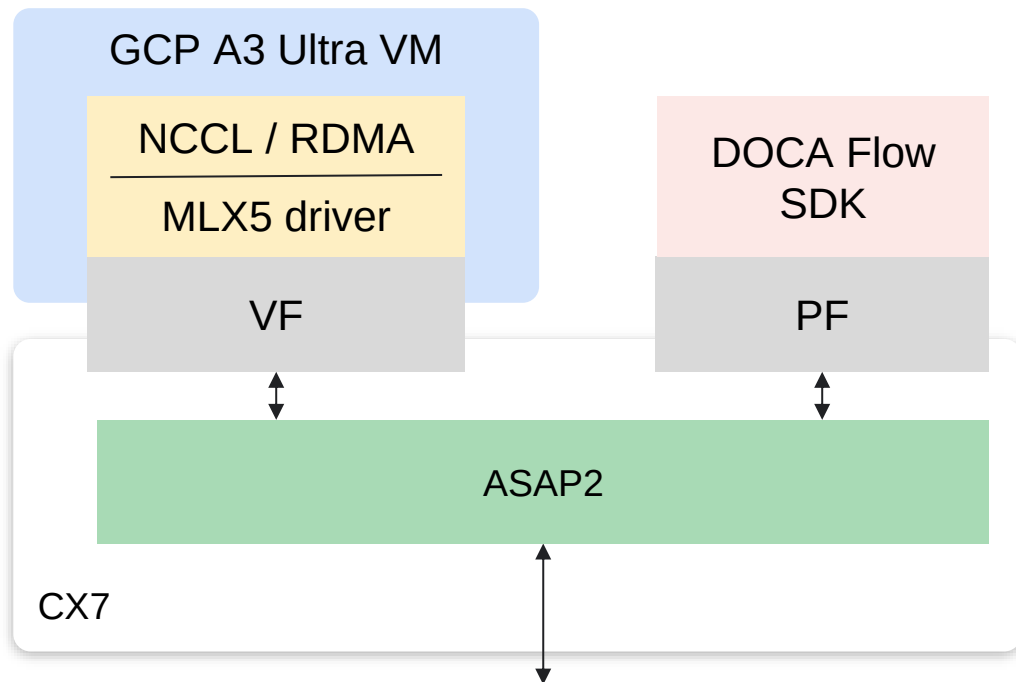
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## DOCA Flow SDK

Works on PF function.

APIs to setup ASAP2 rules.



# DOCA Flow SDK & CX7 NIC

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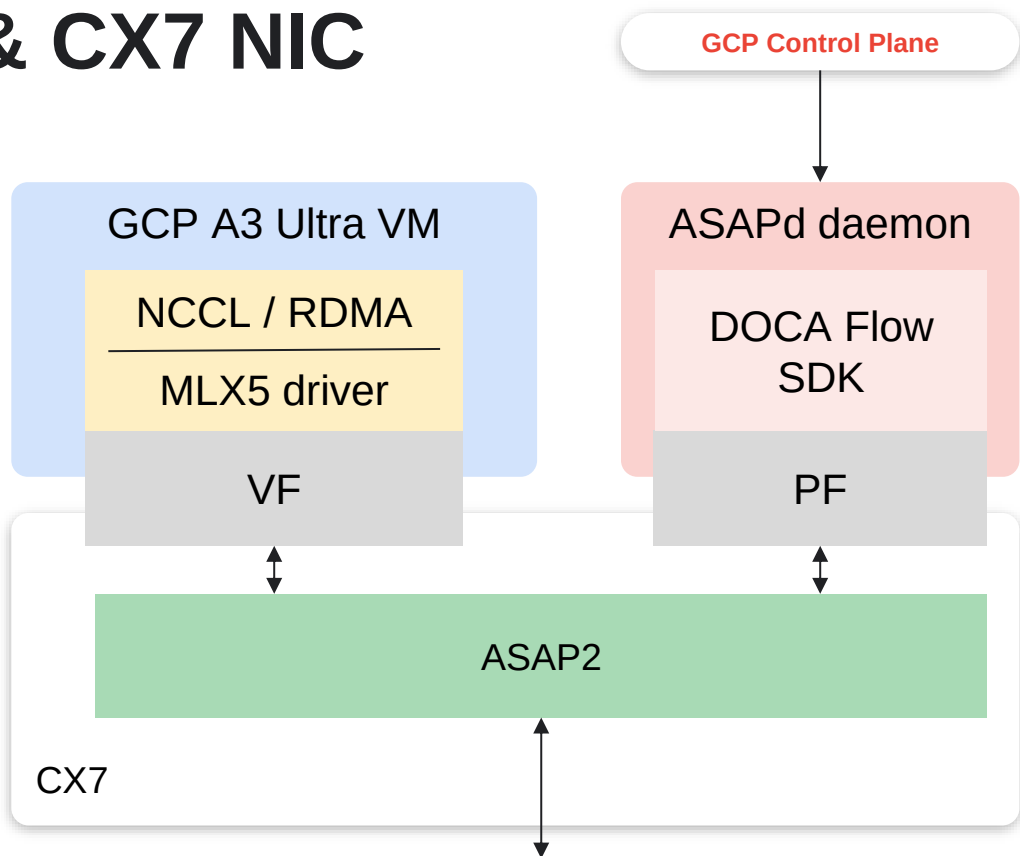
## DOCA Flow SDK

Works on PF function.

APIs to setup ASAP2 rules.

## ASAPd daemon

Convert GCP control plane signal to DOCA Flow SDK packet processing rules.



# Basics of DOCA Flow SDK

## C libraries with APIs to:

Create match/action tables (pipes).  
Insert/remove entries to pipes.

## Supported basic pipe types:

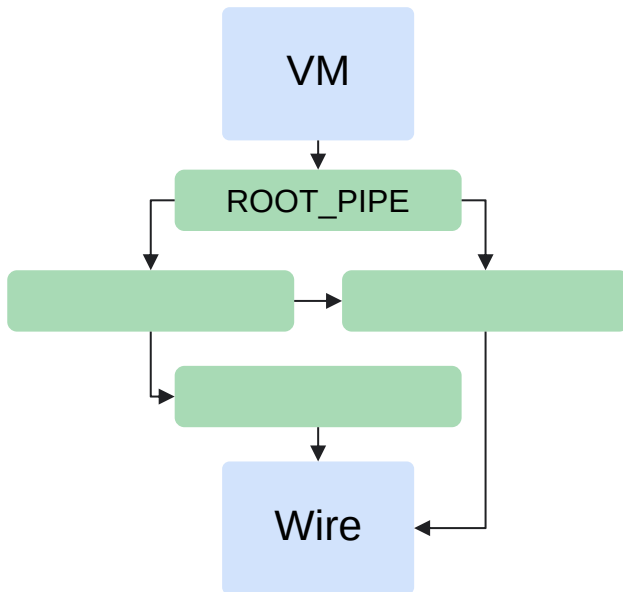
ACL, LPM, Hash, etc.

## Supported basic entry actions:

Modify, mirror, meter packets.  
Forward to a port or to another pipe.

## Forwarding to another pipe allows creation of a packet processing graph.

The first pipe in a graph is a 'root' pipe.

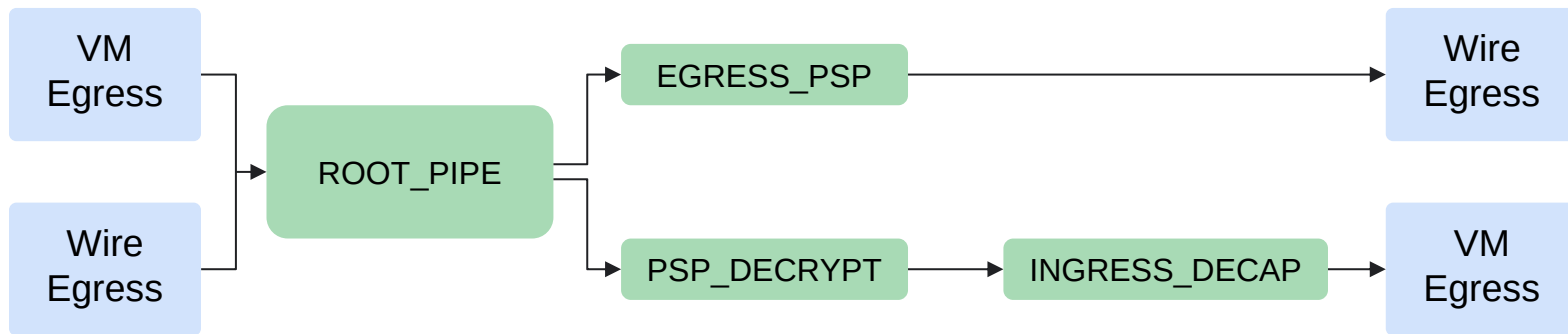


# Basics of DOCA Flow SDK

## Basic Google VPC Overlay App

ROOT\_PIPE: Match [port\_id]  $\Rightarrow$  select To-Wire vs To-VM paths.

- To-Wire: EGRESS\_PSP: Match [3-tuple]  $\Rightarrow$  PSP encapsulation action.
- To-VM: INGRESS\_DECAP: Match [3-tuple, PSP\_SPI]  $\Rightarrow$  forward vs drop

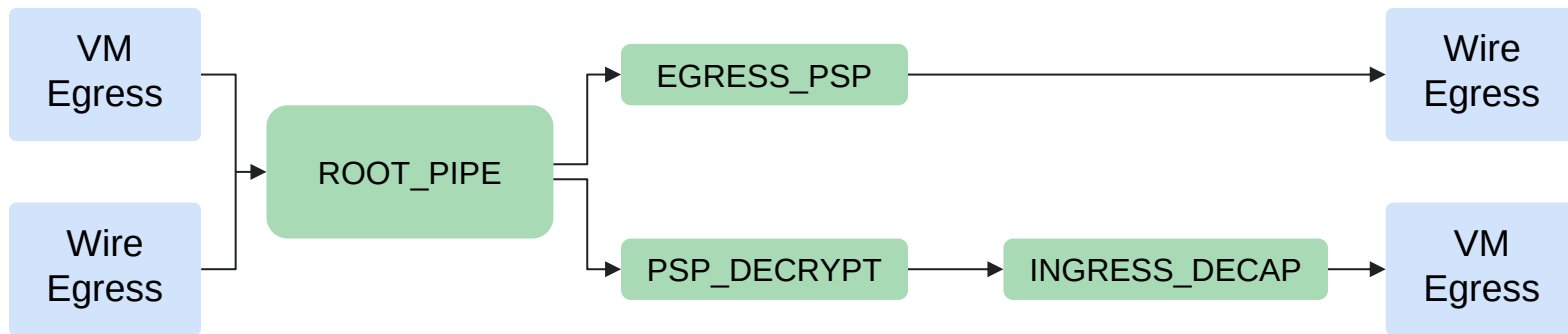


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**How much overhead does this encapsulation + encryption add?**

# CX7 encapsulation and encryption overheads

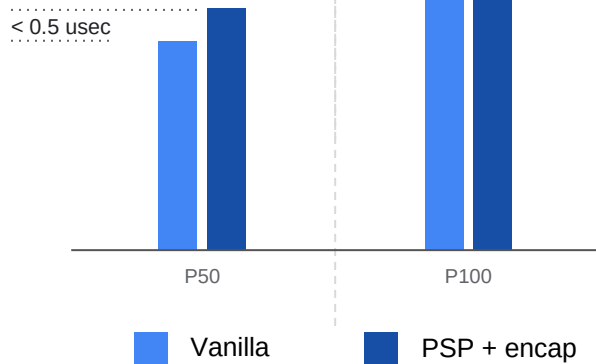
## CX7 PSP encap overheads in OTS NICs:

Example app: [DOCA/applications/psp\\_gateway](https://docs.nvidia.com/doca/architecture/2-7-0/nvidia+doca+psp+gateway+application+guide/index.htm)\*

Wire overhead: ~2.1% IPv6+PSP at 4K frames.

Exceptionally well controlled latency tails.

\*<https://docs.nvidia.com/doca/architecture/2-7-0/nvidia+doca+psp+gateway+application+guide/index.htm>  
|



### Intra-rack latency

DOCA 2.7, ib\_write\_lat, 2B messages, 100K runs.

# CX7 encapsulation and encryption overheads

## CX7 PSP encap overheads in OTS NICs:

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## CX7 PSP experience in GCP

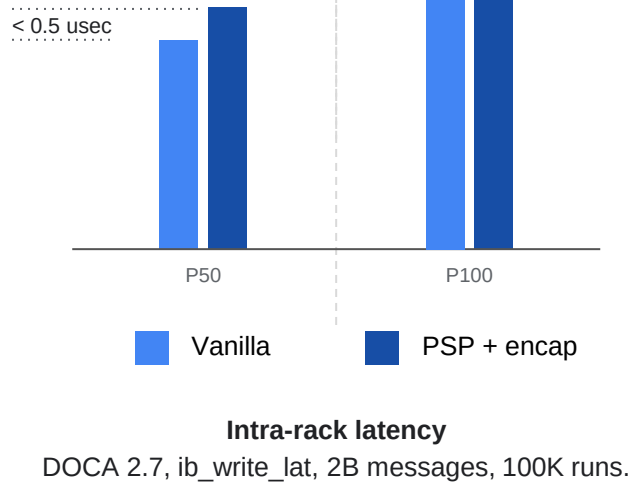
A3-Ultra VM GA in December 2024

First month of use in prod: 100Tbps+ bursts; 200K+ PSP tunnels.

Negligible overhead for higher-level workloads: LLama3 and Mixtrail-8 on GCP A3U CX7 RoCEv2 have almost **identical step times** to NVIDIA's simulated rooflines with CX7 IB.

\*<https://docs.nvidia.com/doca/archive/2-7-0/nvidia+doca+psp+gateway+application+guide/index.htm>

|





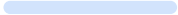
# DOCA use case: Google VPC Traceroute



**Tenant escalation:**  
**“network is broken!”**

Myriads of reasons why network  
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# DOCA use case: Google VPC Traceroute



## Tenant escalation: “network is broken!”

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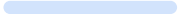


## Network operator’s default tool: packet- level traceroute

**Hypervisor traceroute:** does not exercise VPC processing (e.g., firewall policies).

**In-guest VM traceroute:** Google has no access to customer VMs.

# DOCA use case: Google VPC Traceroute



## Tenant escalation: “network is broken!”

Myriads of reasons why network may appear to be broken.



## Network operator’s default tool: packet- level traceroute

**Hypervisor traceroute:** does not exercise VPC processing (eg. firewall policies).

**In-guest VM traceroute:** Google has no access to customer VMs.



## VPC traceroute requirements for DOCA datapath:

**Inject** probes into the customer’s VPC network **as if** the packets are coming from a guest VM.

**Detect and intercept** trace packets just **before** the delivery to the target VM.

# VPC traceroute with DOCA: Marking trace packets

## PSP header has S and D bits

S: Hint for the receiver to sample.

D: Hint for the receiver to drop.

IFF both bits are set, the effect is to intercept the packet and send it to the control plane.

Paddywhack Security Payload (PSP) Header

```
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Next Header | Hdr Ext Len | R | CryptOffset | S D | version | V | 1 |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                                     Security Parameters Index (SPI)                                     |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                                                                                                                                              |
+               Initialization Vector (IV)               +
|                                                                                                                                              |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                                                                                                                                              |
+               Virtualization Cookie (VC) [Optional]               +
|                                                                                                                                              |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
```

UDP  
header

PSP  
header

PSP payload

PSP  
trailer

# VPC traceroute with DOCA: Marking trace packets

## PSP header has S and D bits

S: Hint for the receiver to sample.

D: Hint for the receiver to drop.

IFF both bits are set, the effect is to intercept the packet and send it to the control plane.

## Pleasant surprise:

The first DOCA PSP API implementation drop had matchers and setters for PSP S and D bits.

Paddywhack Security Payload (PSP) Header

```
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Next Header | Hdr Ext Len | R | CryptOffset | S | D | Version | V | 1 |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Security Parameters Index (SPI) |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Initialization Vector (IV) |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Virtualization Cookie (VC) [Optional] |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
```

UDP  
header

PSP  
header

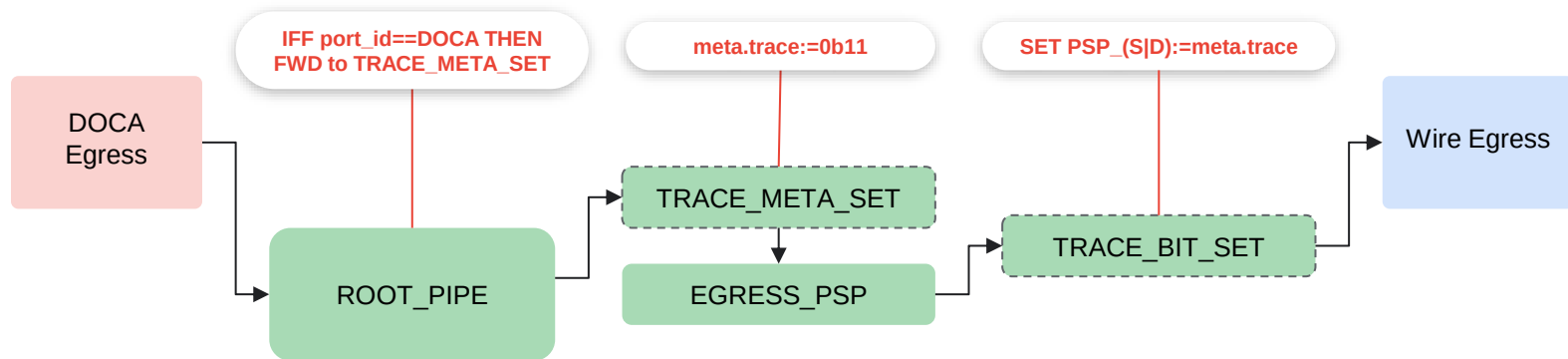
PSP payload

PSP  
trailer

# VPC Traceroute with DOCA

Inject trace packets with DOCA Flow SDK.

Egress: SET PSP\_S|D bits IFF packet is injected from DOCA.



# VPC Traceroute with DOCA in action

CLI and Web UIs.

Mature tool that supports various Google Cloud datapaths.

CX7 DOCA Flow SDK datapath is the latest addition. 

```
valas@valas0:~$ /google/bin/releases/netslo-prod/cloud-traceroute-ng/cloud_traceroute.par --e
nvironment=STAGING --srcVnid=5695825 --dstIp=192.168.6.229 --srcIp=192.168.6.236 --srcEndpoi
ntType=VM --payloadSize=2
Sending Cloud Traceroute 100% |Goooooooooooooooooooooooooooooooooooooooooooooooooooo| Time: 00:00:06

Successful (DROPPED_BY_USPS) - no network failures detected to destination.

Trace events:
- yucbfad-cbh3 (EGRESS_USPS) - UNSPECIFIED
  - (ProcessingEvent: PROBE_INJECTION_STAGE_INJECTED)
- yucbfad-cch3 (INGRESS_USPS) - DROPPED
  - /asapd/sampling_coprocessor/24463385330976485: Packet dropped: Trace request handled

See full results in Cloud Traceroute UI: https://cloud-traceroute-ng-preprod-qual-global.corp.google.com/historicalView?timestampNs=1740435704671234705&traceId=10537466957714029487&recordType=CLOUD\_TRACEROUTES

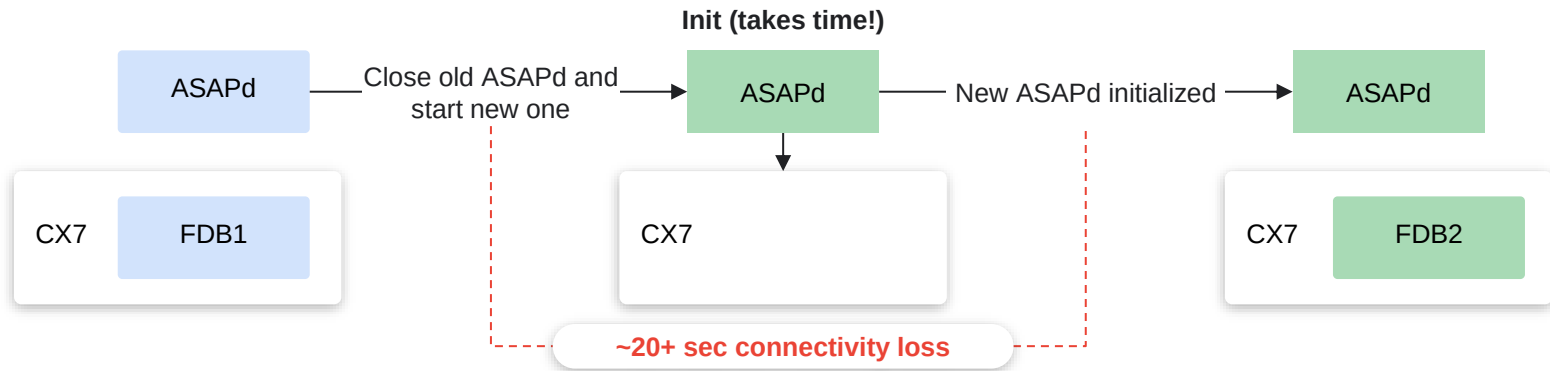
valas@valas0:~$
valas@valas0:~$
```

# DOCA use case: Hitless updates

ASAPd (Google's DOCA daemon) needs updates.

**Restarting a running DOCA Flow SDK will cause network connectivity loss.**

Process close → FD close → NIC resource clean up → ASAP2 FlowDB (FDB) state cleanup.



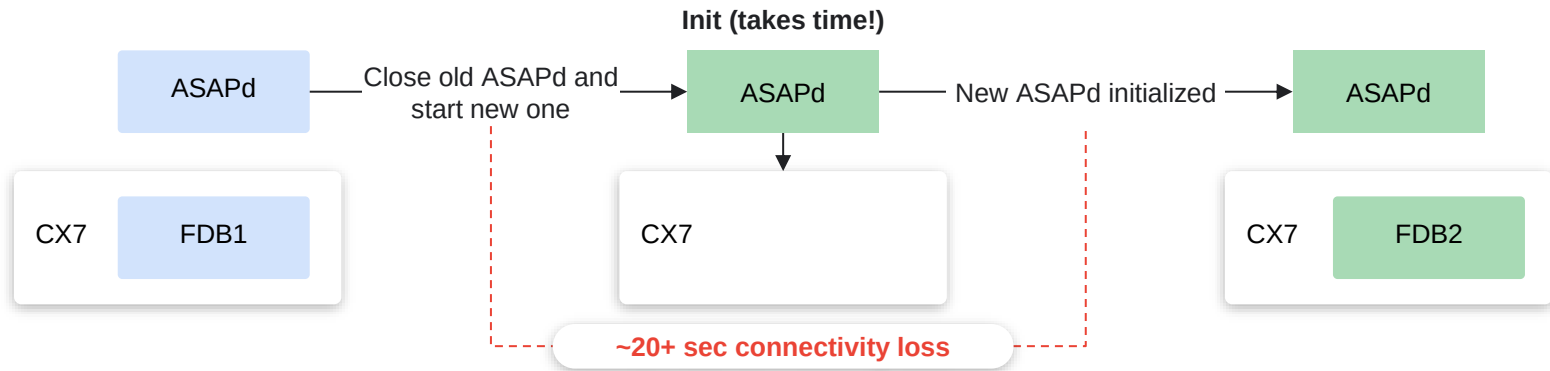


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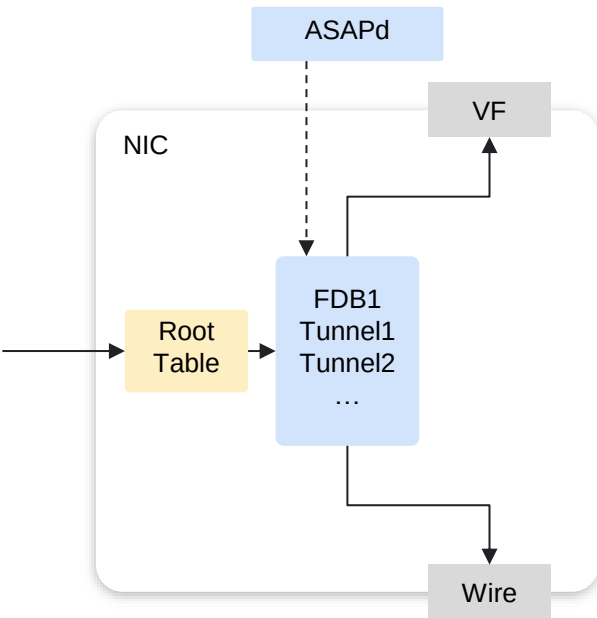


**20 sec datapath blackout is unacceptable.**

Connectivity loss → RDMA QP timeout → training job dies → restart from checkpoint.

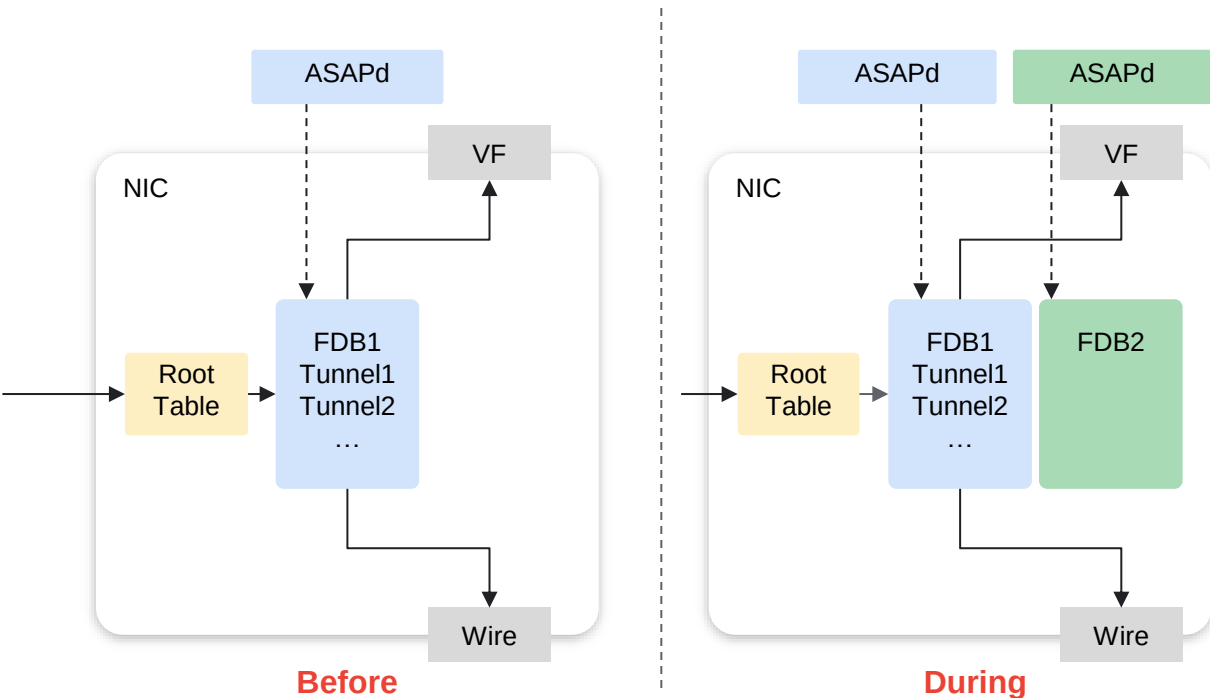
Rolling software updates ensure prolonged pain on large training clusters.

# Hitless update process

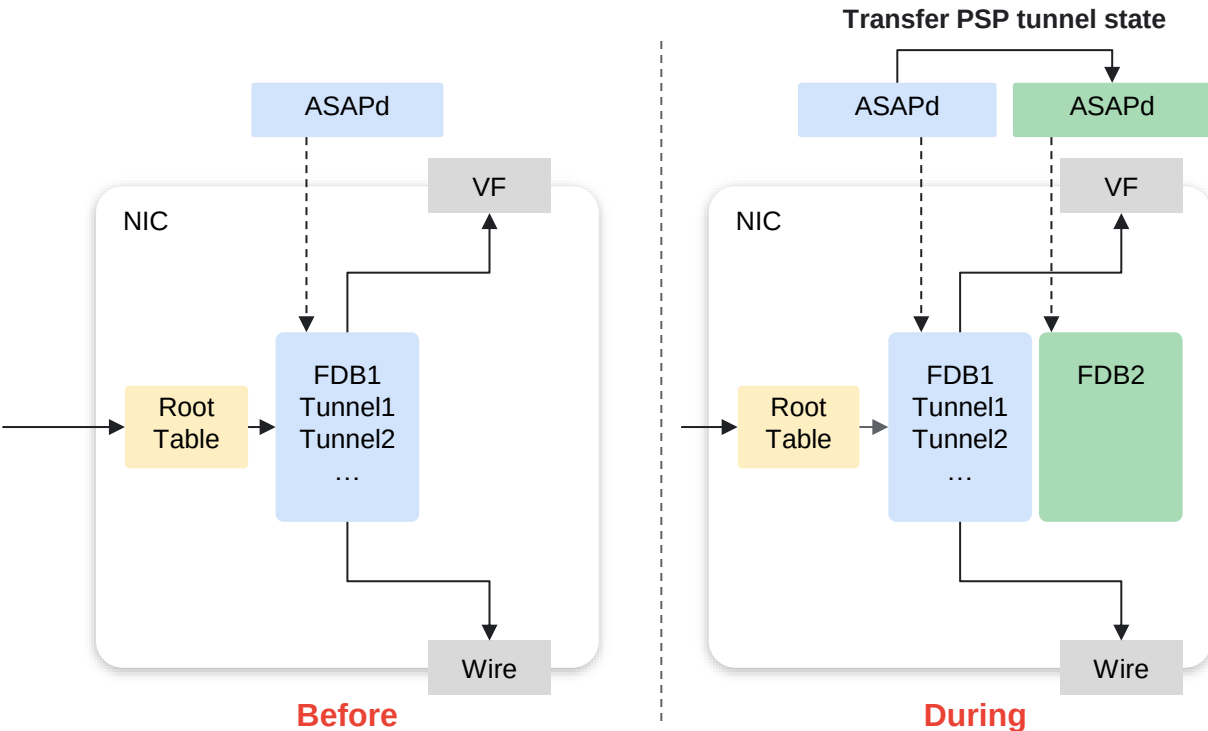


**Before**

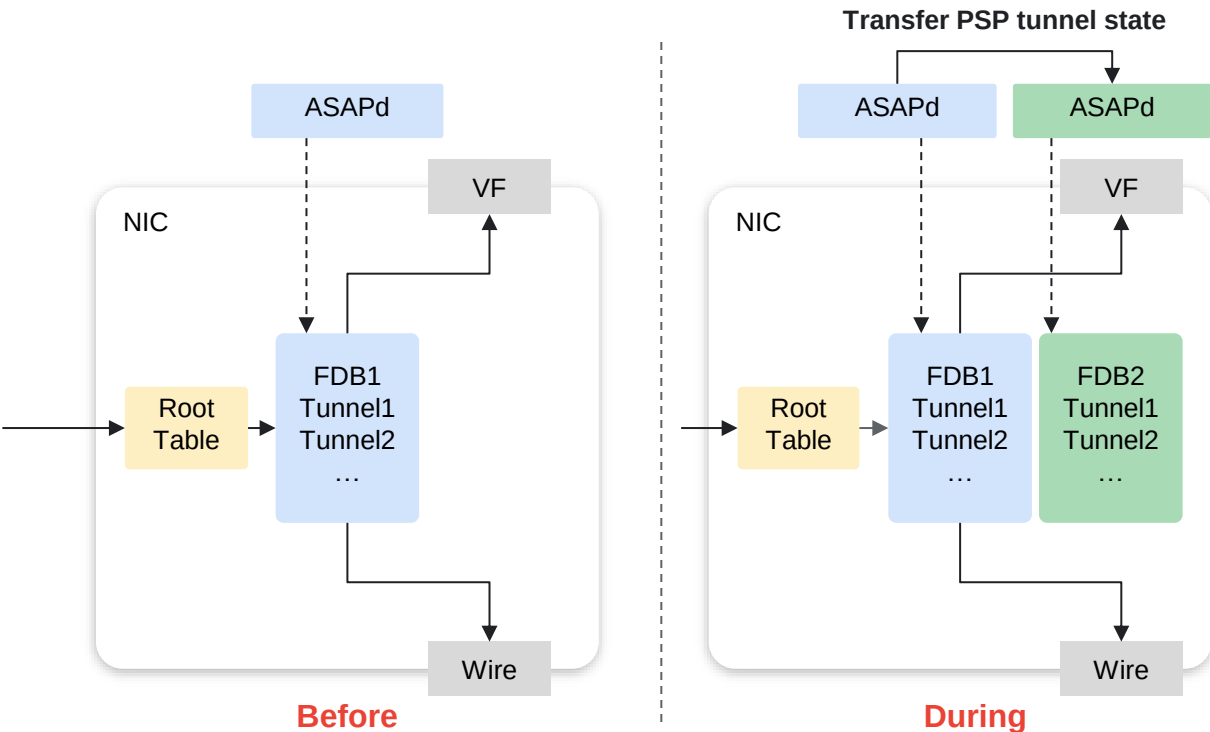
# Hitless update process



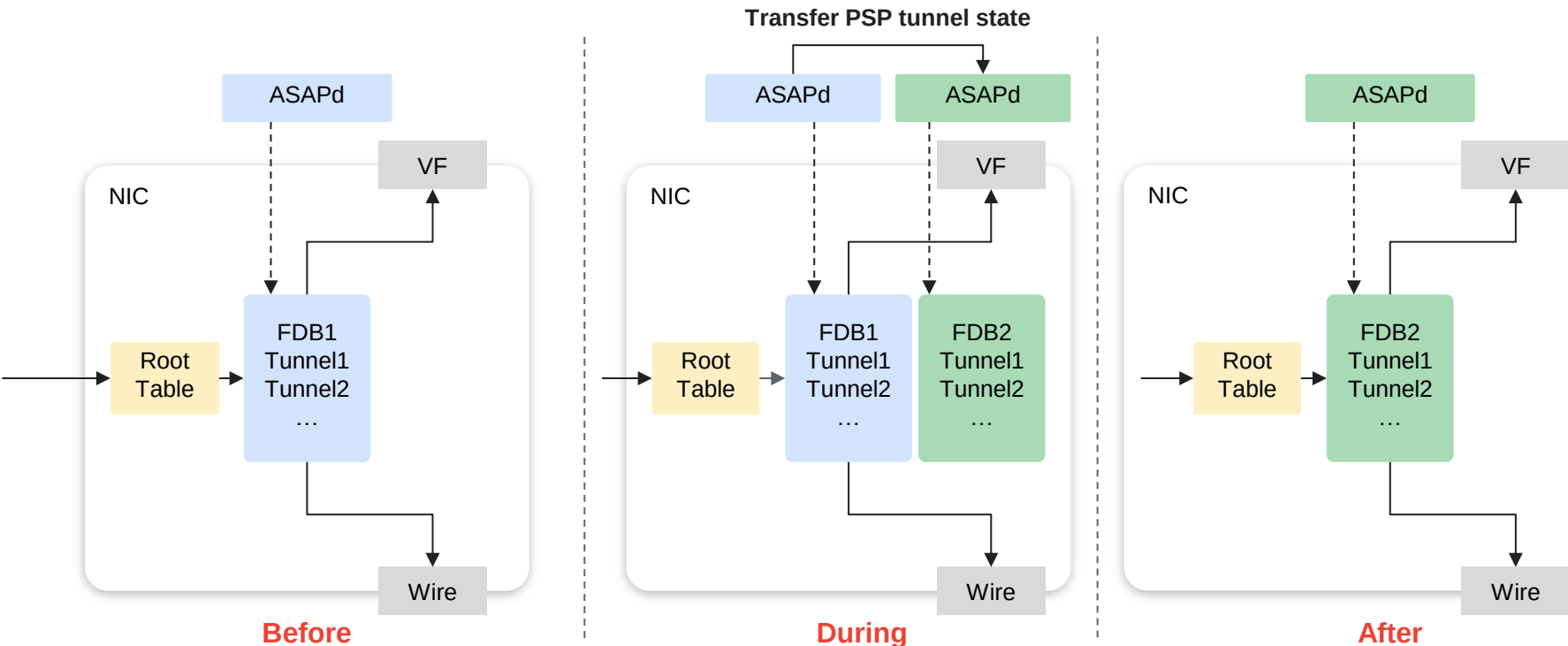
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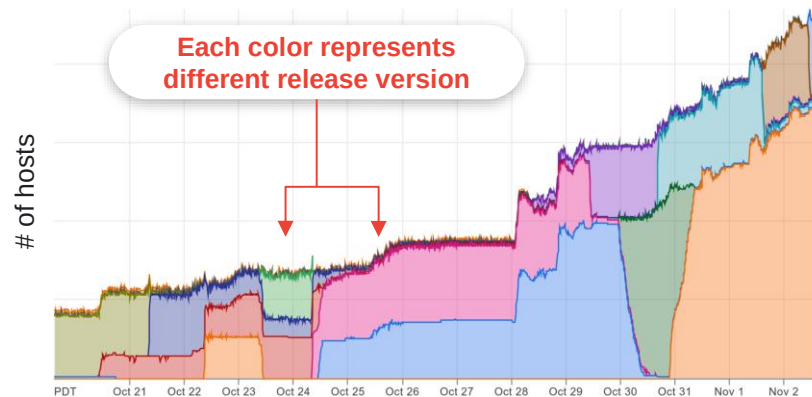
# Hitless updates experience in Google Cloud

## Cadence:

Daily build; at least weekly release to staging; at least monthly to production.

Over the Nov-Dec-Jan period: 48 staging releases, 4 prod releases, zero disruption to AI/ML workloads.

**ASAPd versions snapshot over two weeks in October 2024.**



# DOCA @ Google



## CX7 NICs power Google Cloud A3 Ultra VMs

CX7 RDMA networking stack is familiar to tenants coming from on-prem.

CX7 delivers PSP integrity and confidentiality for RoCE traffic at low cost.



## DOCA powers CX7 NICs in Google Cloud

Well-polished and stable APIs make it easy to develop advanced features.

CSP-grade features, such as rich telemetry and hitless upgrades.



## DOCA delivers agility to Google Cloud

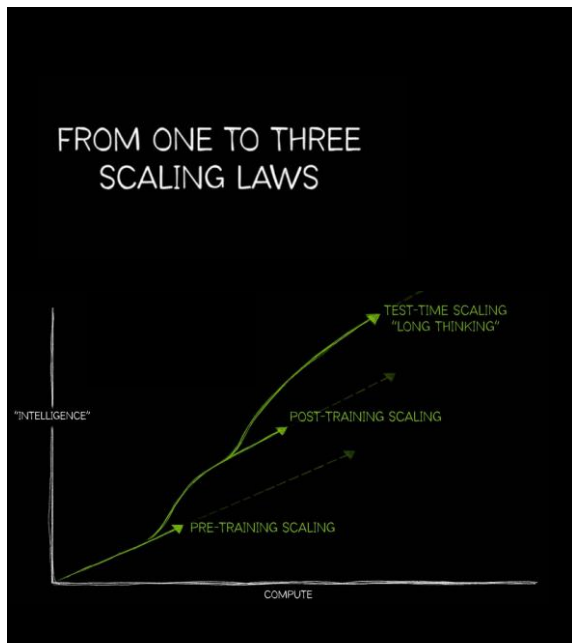
PSP encryption and hitless update features developed in a matter of months.



The background features a series of overlapping, curved, light green bands that create a sense of depth and movement, resembling a stylized staircase or a series of parallel planes. The bands are set against a white background on the left side.

**DOCA 3.0**

# Key Innovations Shaping the Future of AI



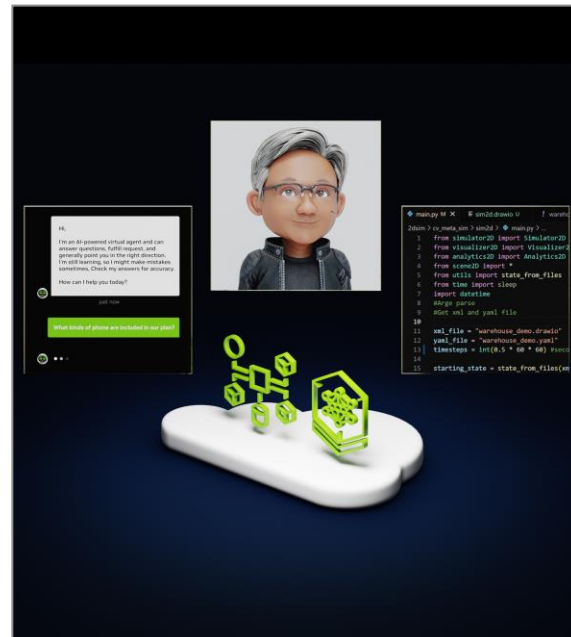
## Test-Time Scaling

Applying more compute at inference time to enhance accuracy and model performance



## Sovereign AI

Driving localized, secure AI infrastructure, minimizing external reliance, and ensuring data control



## AI Reasoning

Using sophisticated reasoning and iterative planning to autonomously solve complex, multi-step problems

# Announcing NVIDIA DOCA 3.0 to Power the Next Frontier of AI Platforms



## Hyperscale GPU Computing

Scale multi-tenant AI factories beyond 100K GPUs with unmatched performance and efficiency



## Robust Threat Detection

Immediately detect and stop runtime attacks in AI workloads, on any compute



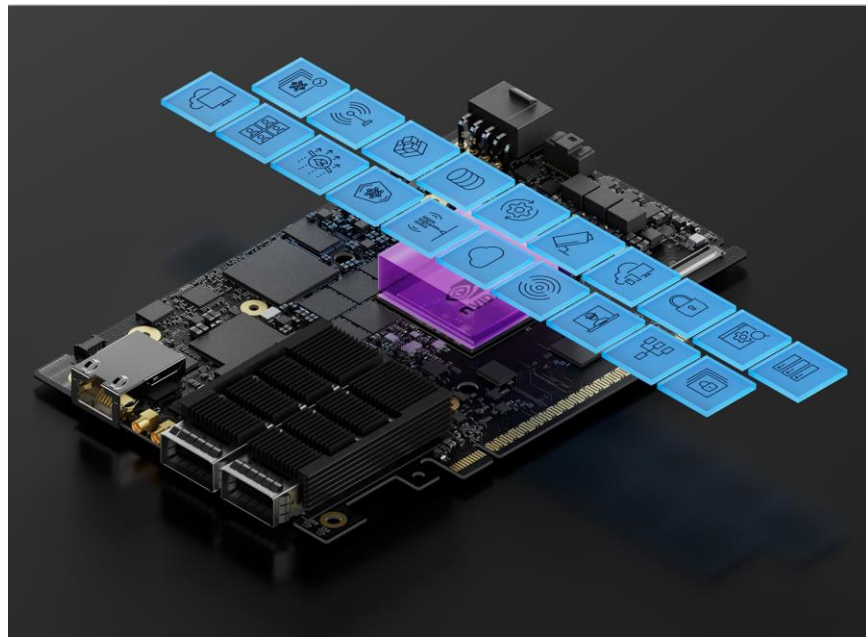
## Intelligent Data Platform

Accelerate data processing for the next wave AI workloads with high-performance networking



## Infrastructure Service Orchestration

Seamlessly deploy and manage DPU-powered networking, storage, and security acceleration services



# Announcing NVIDIA DOCA 3.0 to Power the Next Frontier of AI Platforms

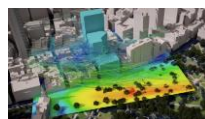
## NVIDIA AI



Generative AI



Data Science



Inference



Vision AI



Cybersecurity AI

Optimized Framework for AI  
Data Center Infrastructure

Seamless Integration

In-Depth Security & Visibility

## DOCA 3.0

Infrastructure Services for AI Factories

Service Orchestration

Intelligent Data Fabric

Threat Detection

Programmable Congestion  
Control

Crypto

Telemetry

Software Defined Networking

Storage Acceleration

Security Groups

NVIDIA BlueField DPU, BlueField & ConnectX SuperNICs



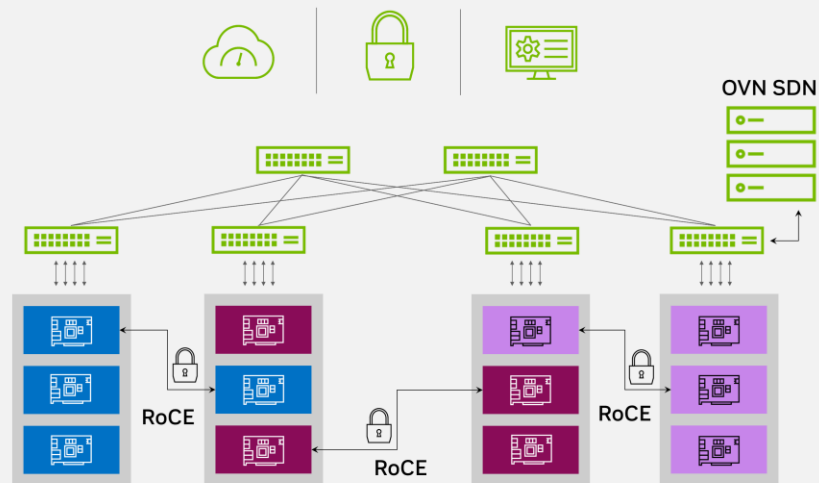
# OVS-DOCA Service

Accelerate multi-tenant VPC networking and enforce security policies

- Accelerated by DOCA Flow with latest ASAP<sup>2</sup> steering technology
- Encrypted S2S and fabric for multi-tenant security (PSP, IPsec)
- Drop-in replacement version for OVS (OVN compatible)
- Unmatched scale and connection tracking performance
- Unified service for AI and user-to-cloud networks (E-W, N-S)

Learn more in this GTC session:

**Power Sustainable AI Clouds With Advanced Networking [S74316]**



VPC for Tenant A

VPC for Tenant B

VPC for Tenant C

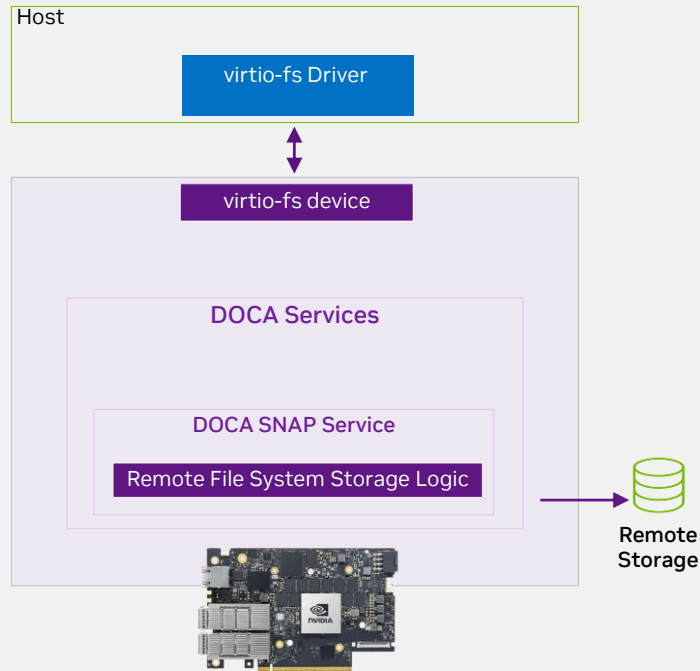
# DOCA SNAP VirtIO-FS

Accelerate, scale and secure the distributed network file system

- DOCA service for high throughput and parallel file system storage
- Cloud-scale distributed file system storage enablement in AI compute servers
- Secured file system through isolation and policy enforcements
- Emulate host local file system semantics using virtio-fs driver
- Accelerating file system performance and offload remote file system logic to the DPU

Learn more in this GTC session:

[Storage Innovations for AI Workloads \[S72329\]](#)



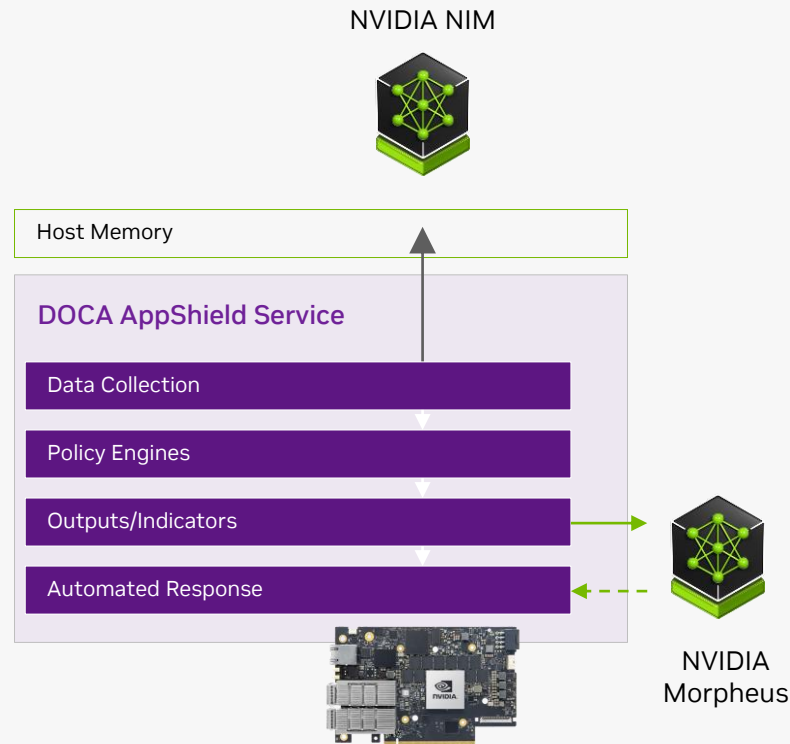
# DOCA AppShield for NIM Security

## Detecting Runtime Attacks in NVIDIA NIMs

- Focused, purpose-built for NIMs & AI workloads
- Immediately detects and stops runtime attacks to minimize potential impact and risk
- Gain real-time workload visibility at the network, host, application level and File System by analyzing specific regions of volatile memory
- Agentless deployment, ensuring peak AI performance
- Flow based automated response enforcement

Learn more in this GTC session:

**Power the Next Wave of AI with Real-Time Runtime Security [S71642]**



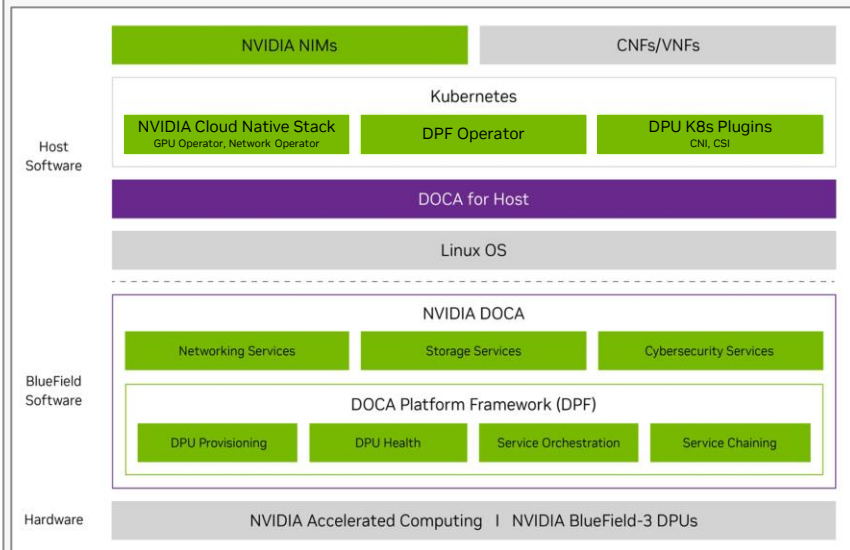
# DOCA Platform Framework (DPF)

## BlueField DPU Provisioning & Orchestration Framework for Cloud Data Centers

- Orchestrate DPU based infrastructure acceleration services
- Boost performance by seamless offload of networking, storage & security
- Simplify DOCA and 3<sup>rd</sup> party services deployment & chaining
- Cluster-wide visibility, control and life-cycle management for BlueField DPUs
- Integrate with K8s and leading ISVs

Learn more in this GTC session:

[Enable AI-Native Networking for Telcos with Kubernetes \[S72993\]](#)





# DOCA 3.0 Accelerates and Secure NIMs & AI workloads

